

TELECOM CUSTOMER CHURN PREDICTION

Problem Statement

Customer churn is a critical issue in the telecommunications industry, where retaining existing customers is often more cost-effective than acquiring new ones. Telecom companies face significant revenue loss when customers discontinue their services. Therefore, predicting customer churn is essential for developing effective retention strategies.

This project focuses on building a machine learning-based predictive system to identify customers who are likely to leave the telecom service. By analyzing customer demographic details, subscribed services, and account-related information, the project aims to help businesses proactively target at-risk customers and reduce churn.

Objectives:

- Predict whether a customer is likely to churn
- Analyze customer behavior using available data
- Compare machine learning models for churn prediction
- Support customer retention programs through predictive insights

Dataset Used

Dataset Name:

IBM Telecom Customer Churn Dataset

Dataset Overview:

- Total Records: 7043 customers
- Total Features: 21 columns
- Target Variable: Churn

Dataset Context:

The dataset is designed to analyze customer behavior and predict retention.

Data Contents:

Customer Churn Information:

- Churn (whether the customer left within the last month)

Services Signed Up:

- Phone service
- Multiple lines
- Internet service
- Online security
- Online backup
- Device protection
- Tech support
- Streaming TV
- Streaming movies

Customer Account Information:

- Tenure
- Contract type
- Payment method
- Paperless billing
- Monthly charges
- Total charges

Demographic Information:

- Gender
- Senior citizen status
- Partner status
- Dependents

Data Characteristics:

- Structured tabular dataset
- Supervised classification problem
- Binary target variable (Churn: Yes/No)

Approach / Methodology

Data Preparation:

- Used IBM Telecom Customer Churn dataset
- Selected customer attributes as input features
- Used Churn column as target variable
- Divided data into training and testing sets
- Applied machine learning classification algorithms

Machine Learning Models Used:

Decision Tree Classifier:

Decision Tree is a supervised learning algorithm that classifies data by splitting it into branches based on feature importance.

Random Forest Classifier:

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Evaluation Methods:

- Accuracy Score
- Confusion Matrix
- 5-Fold Cross-validation

Results and Evaluation

Decision Tree Results

Accuracy Score:

79.10%

Confusion Matrix:

[[938, 94],
[199, 171]]

Cross-validation Scores:

[0.77272727, 0.78253119, 0.78966132, 0.79393399, 0.7618198]

Average Cross-validation Accuracy:

78.01%

Performance Analysis:

- Correctly predicted many non-churn customers
- Moderate churn detection performance
- Some churn customers were misclassified
- Stable model performance across validation folds

Random Forest Results

Accuracy Score:

80.10%

Confusion Matrix:

[[937, 95],
[184, 186]]

Performance Analysis:

- Slightly higher accuracy than Decision Tree
- Better prediction of churn customers
- Lower false negatives
- More balanced overall performance

Comparative Analysis

Metric	Decision Tree	Random Forest
Accuracy Score	79.10%	80.10%
True Positives	171	186
False Negatives	199	184

Conclusion

This project successfully developed machine learning models to predict telecom customer churn using customer demographic, service, and account-related data.

Final Performance:

- Decision Tree Accuracy: 79.10%
- Random Forest Accuracy: 80.10%

Best Model:

Random Forest Classifier

Reason:

Random Forest performed better due to:

- Higher accuracy
- Better churn customer prediction
- Lower false negatives
- Improved reliability

Thus, Random Forest is more suitable for practical telecom customer retention systems.